$Q \in M_{m,n}(\mathbb{R})$ $\quad \underline{Q^t Q = I_n} \quad \not\Leftrightarrow \quad QQ^t = I_m$!
$(n,1)(n,m)$

$Q = (q_1 | \dots | q_n) \qquad q_1 \dots q_n \in \mathbb{R}^m$

$$Q^t Q = \begin{pmatrix} \frac{q_1^t}{} \\ \frac{q_2^t}{} \\ \vdots \\ \frac{q_n^t}{} \end{pmatrix} (q_1 | \dots | q_n) = (q_i^t q_j)_{i,j} =$$

$$= (\langle q_i, q_j \rangle)_{i,j=1,n} \overset{?}{=} I_n \quad \Leftrightarrow$$

$$\Leftrightarrow \langle q_i, q_j \rangle = \begin{cases} 1, & i = j \\ 0, & \text{in rest} \end{cases}$$

$\Leftrightarrow$ $q_1 \dots q_n$ vectori ortonormati în $\mathbb{R}^m$

$(\Rightarrow \quad n \leq m)$

$n < m$	sau	$n = m$

$1 \Leftrightarrow 2$ (mai mult)

$1 \Rightarrow 3 \Rightarrow 4 \Rightarrow 2 \Rightarrow 1$

$1 \Rightarrow 3$ $\quad \langle Qx, Qy \rangle = (Qx)^t Qy =$

$= x^t \underbrace{Q^t Q}_{I_n} y = x^t I_n y = x^t y = \langle x, y \rangle$

$3 \Rightarrow 2$ $\quad Qe_i = q_i \quad \forall i = \overline{1,n}$

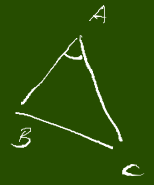
$$Q = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \dots & \dots & \dots \\ a_{n1} & \dots & a_{nn} \end{pmatrix} \qquad e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \qquad Qe_1 = \begin{pmatrix} a_{11} \\ \vdots \\ a_{n1} \end{pmatrix} = q_1$$

$\Rightarrow \quad \langle q_i, q_j \rangle = \langle Qe_i, Qe_j \rangle \overset{3)}{=} \langle e_i, e_j \rangle = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$

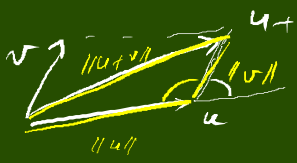
$\Rightarrow q_1 \dots q_n$ vectori ortonormati.

$3 \Rightarrow 4$ $\quad \|x\| = \sqrt{\langle x, x \rangle}$

$4 \Rightarrow 3$ $\quad \langle x, y \rangle =$



$$\cos A = \frac{AB^2 + AC^2 - BC^2}{2AB \cdot AC}$$



$$-\frac{\|u\|^2+\|v\|^2-\|u+v\|^2}{2\|u\|\cdot\|v\|}=\frac{\langle u,v\rangle}{\|u\|\cdot\|v\|}$$

$u,v\neq 0$

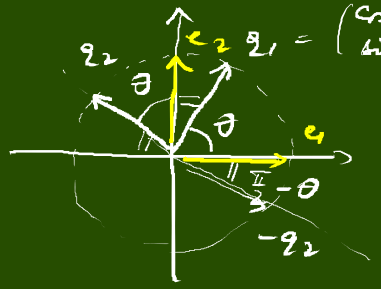
$$\langle u,v\rangle=\frac{1}{2}\left(\|u+v\|^2-\|u\|^2-\|v\|^2\right)$$

$$\|u+v\|^2=\langle u+v,u+v\rangle=\|u\|^2+\|v\|^2+2\langle u,v\rangle$$

exemples : 1) $n=2=m$

$$Q=\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}\qquad \theta\in\mathbb{R}$$

$$Q^tQ=I_2\qquad q_1=\begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix},\ q_2=\begin{pmatrix}-\sin\theta\\ \cos\theta\end{pmatrix}\in\mathbb{R}^2$$



rotation de angle θ

$$\begin{cases} u\longrightarrow -q_2 & \text{rotation } \frac{\pi}{2}-\theta\\ e_2\longrightarrow q_1\end{cases}$$

$\longrightarrow \quad Q' = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$ trasf. ?

$\begin{matrix} \| & \| \\ q_1 & -q_2 \end{matrix}$

$\begin{cases} e_1 \longrightarrow q_1 \\ e_2 \longrightarrow -q_2 \end{cases}$

2) $m > n \quad (m = n)$ $\qquad Q = \begin{pmatrix} 2/3 & 2/3 \\ 1/3 & -2/3 \\ 2/3 & -1/3 \end{pmatrix}$

Q matrice ortogonale $\quad Q^t Q = I_2$ ~~I_3~~

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1) $A = P \cdot D \cdot P^{-1}$ $\qquad P$ invers., D diagonale

2) $A = Q \cdot R$

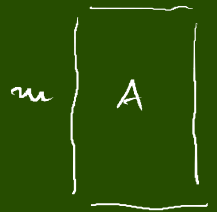
$A = (v_1 | \dots | v_n)$

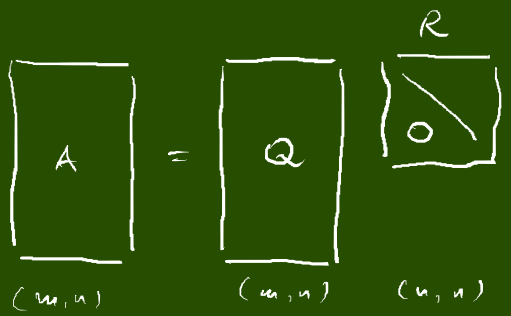
$\operatorname{rang} A = n \Leftrightarrow v_1 \dots v_n$ lin. indep. $\Big| \Rightarrow$

$\operatorname{Im}(A) = \operatorname{Sp}\{v_1 \dots v_n\}$

$\Rightarrow \{v_1 \dots v_n\}$ base in $\operatorname{Im}(A)$ $\quad \dim_{\mathbb{R}} \operatorname{Im}(A) = n$

$\Rightarrow \quad n \leq m$

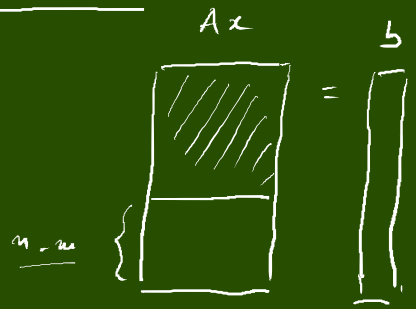




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1) $Ax = b$

$A = QR$



$\operatorname{rang} A = n \iff$ unicitatea sol.

$\operatorname{rang} A = n \iff \exists$ sol.

$QRx = b \quad / Q^{-1}$

$Rx = Q^{-1}b$



2) $\sigma(A)$

$A = QR$

$A_1 = RQ$

$A_1 = Q_1 R_1$

$A_2 = R_1 Q_1$

$\lim_{n \to \infty} A_n =$

3) $\begin{cases} x + y = 1 \\ x = 1 \\ y = 1 \end{cases}$

$\operatorname{rang} A = n \qquad Q = ? \quad R = ?$

$\Downarrow$

$v_1, \dots v_n$ lin indep $\qquad A = (v_1 | \dots | v_n)$

$\Downarrow$

$v_1, \dots, v_n$ bază în $\mathrm{Im}(A)$

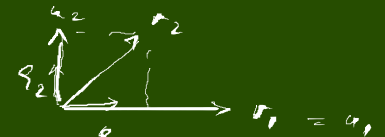
$\Downarrow$ Gram-Schmidt

$g_1, \dots, g_n$ bază ortonormată în $\mathrm{Im}(A)$

$\Downarrow$

$Q = (g_1 | \dots | g_n)$ matrice ortogonală.

$$\begin{cases} v_1 = \langle v_1, g_1 \rangle g_1 \\ v_2 = \langle v_2, g_1 \rangle g_1 + \langle v_2, g_2 \rangle g_2 \\ \vdots \\ v_n = \langle v_n, g_1 \rangle g_1 + \dots + \langle v_n, g_n \rangle g_n \end{cases}$$



$g_1 \neq 0$

$$\Rightarrow A = (v_1 | \dots | v_n) = \underbrace{(g_1 | \dots | g_n)}_{Q} \cdot \begin{pmatrix} \langle v_1, g_1 \rangle & \langle v_2, g_1 \rangle & \dots & \langle v_n, g_1 \rangle \\ 0 & \langle v_2, g_2 \rangle \neq 0 & \dots & \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & \langle v_n, g_n \rangle \neq 0 \end{pmatrix}$$

Exemplu. $A = \begin{pmatrix} 2 & 6 \\ 1 & -3 \\ 2 & 0 \end{pmatrix}$ $\operatorname{rang} A = 2$ $\Rightarrow$ R inversabilă.

u_1 u_2

cursul precedent $\Rightarrow$ (alg GS)

$q_1 = \begin{pmatrix} 2/3 \\ 1/3 \\ 2/3 \end{pmatrix}$, $q_2 = \begin{pmatrix} 2/3 \\ -2/3 \\ -1/3 \end{pmatrix}$

ortonormată, bază în $\operatorname{Im}(A)$ ortonormată

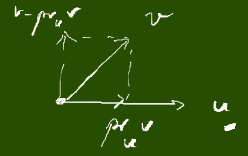
$\Rightarrow Q = (q_1 | q_2) = \begin{pmatrix} 2/3 & 2/3 \\ 1/3 & -2/3 \\ 2/3 & -1/3 \end{pmatrix}$

$A = QR$

$(3,2)\ (3,2)\ (2,2)$

$R = \begin{pmatrix} \langle u_1, q_1 \rangle & \langle u_2, q_1 \rangle \\ 0 & \langle u_2, q_2 \rangle \end{pmatrix} = \begin{pmatrix} 3 & 3 \\ 0 & 6 \end{pmatrix}$

$\Rightarrow A = QR$.



U

$S = \{0\}$ $\quad S^\perp = \{v \in V \mid \langle v, 0 \rangle = 0\}$

$= V$

$S = V$ $\quad S^\perp = \{v \in V \mid \langle v, u \rangle = 0 \ \underbrace{\forall u \in V}_{u \in V}\}$

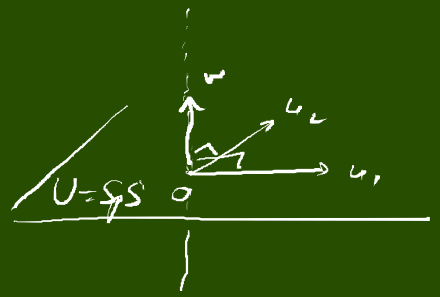
$= \{0\}$

$S = \{u_1, u_2\}$ $\quad S^\perp = ?$

$u_1 \neq 0, u_2 \neq 0$

u_1, u_2 lin. indep

$v \perp u_1$, $v \perp u_2$ $\Rightarrow v \perp U$



$S^\perp$

$U^\perp = Sp\, S^\perp$

S base

exemplu $U = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \mid x + 2y - 3z = 0 \right\}$

$U^\perp = ? \quad = \ker \underbrace{(1 \;\; 2 \;\; -3)}_{A}$

$x + 2y - 3z = 0 \qquad x = -2y + 3z$

$U = \left\{ \begin{pmatrix} -2y+3z \\ y \\ z \end{pmatrix} \mid y, z \in \mathbb{R} \right\} = Sp \left\{ \underbrace{\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}}_{u_1}, \underbrace{\begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}}_{u_2} \right\}$ bază în U

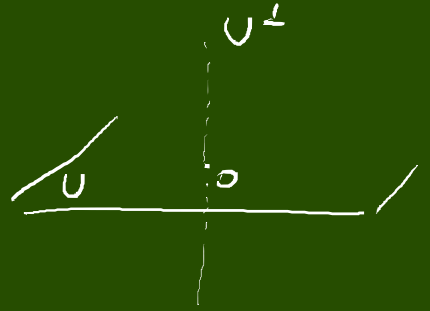
$U^\perp = \{u_1, u_2\}^\perp$

$v = ? \qquad \langle v, u_1 \rangle = 0 \qquad \langle v, u_2 \rangle = 0$

$v = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \qquad -2x + y = 0 \qquad y = 2x$

$3x + z = 0 \qquad z = -3x \qquad A^t$

$\Rightarrow U^\perp = \left\{ \begin{pmatrix} x \\ 2x \\ -3x \end{pmatrix} \mid x \in \mathbb{R} \right\} = Sp \left\{ \overbrace{\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}}^{} \right\} = \ker(A^t)$



$$\begin{cases} U \cap U^\perp = \{0\} \\ U \oplus U^\perp = \mathbb{R}^3 \end{cases}$$

$U \cap U^\perp \ni v$

$v \in U$

$v \in U^\perp \Rightarrow \langle v, u \rangle = 0 \quad \forall u \in U$

pt $u = v \Rightarrow \langle v, v \rangle = 0 \Rightarrow v = 0$

$\Rightarrow U \cap U^\perp = \{0\}$

$U \subseteq (U^\perp)^\perp$

$\supseteq$ " de V finit dimensional

$\ker(A)^\perp = \operatorname{Im}(A^t)$

$\operatorname{Im}(A)^\perp = \ker(A^t)$

V $\quad U \subseteq V \quad v \in V$

$\Rightarrow \exists!\ w_1 \in U,\ w_2 \in U^\perp.\ v = w_1 + w_2.$

Unicitatea $\quad U \cap U^\perp = \{0\}$

$v = w_1 + w_2 = z_1 + z_2 \qquad w_1, z_1 \in U,\ w_2, z_2 \in U^\perp$

$\Downarrow$

$w_1 - z_1 = z_2 - w_2 \in U \cap U^\perp = \{0\}$

$w_1 - z_1 \in U \qquad z_2 - w_2 \in U^\perp$

$\Rightarrow w_1 - z_1 = z_2 - w_2 = 0 \Rightarrow w_1 = z_1,\ w_2 = z_2$

Existența : Fie $\underbrace{e_1, \dots e_n}$ bază ortonormată în U

$v = w_1 + w_2 \qquad w_1 \in U \quad w_2 \in U^\perp$

$w_1 = \alpha_1 g_1 + \dots + \alpha_n g_n \qquad \left(\alpha_i = \langle w_1, g_i \rangle\right)$

$w_2 = v - (x)_1 = v - \sum \alpha_i g_i \in U^\perp$

$\Rightarrow \langle w_2, g_i \rangle = 0 \quad \forall i = \overline{1,n}$

$0 = \langle w_2, g_1 \rangle = \langle v - (\alpha_1 g_1 + \dots + \alpha_n g_n), g_1 \rangle =$

$= \langle v, g_1 \rangle - \alpha_1 \underbrace{\langle g_1, g_1 \rangle}_{1} - \alpha_2 \underbrace{\langle g_2, g_1 \rangle}_{0} - \dots - \alpha_n \underbrace{\langle g_n, g_1 \rangle}_{0}$

$\Rightarrow \alpha_1 = \langle v, g_1 \rangle \quad \dots \quad \alpha_n = \langle v, g_n \rangle$

$\Rightarrow \begin{cases} w_1 = \sum_{i=1}^{n} \langle v, g_i \rangle g_i = pr_U v \\ w_2 = v - (x)_1 = pr_{U^\perp} v \end{cases}$

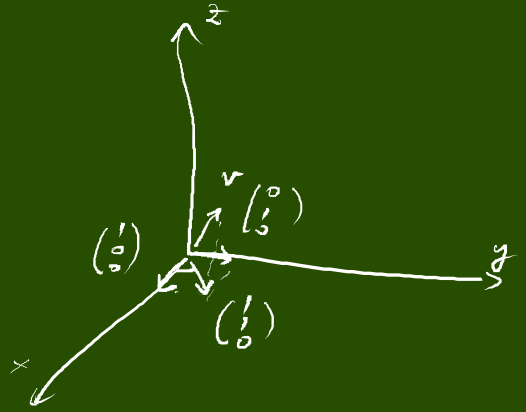
$U \qquad \{u_1 \dots u_n\}$ bază nu neapărat ortogonală (ortonormată)

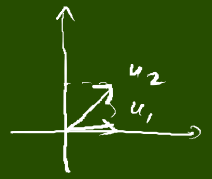
$\Rightarrow pr_U v \neq \sum pr_{u_i} v \quad !!!$

$\mathbb{R}^3$ $\quad U = \left\{ \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} \mid x, y \in \mathbb{R} \right\}$ $\quad xOy$

$u_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $u_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

baza în U





$v = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ $\quad \begin{pmatrix} x \\ y \\ z \end{pmatrix} \xrightarrow{xOy} \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$

$\operatorname{pr}_U v = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = u_2$!

$\operatorname{pr}_U v \neq \operatorname{pr}_{U_1} v + \operatorname{pr}_{U_2} v$!

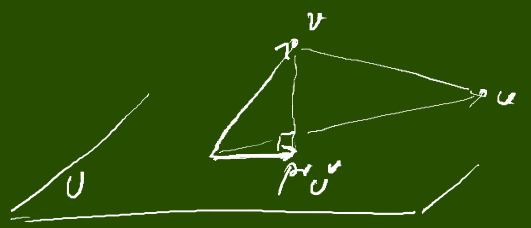
$= \frac{\langle v, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 + \frac{\langle u_1, u_2 \rangle}{\langle u_2, u_2 \rangle} u_2 = \frac{1}{1} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \frac{1}{1} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

$$\begin{cases} x+y=1 \\ x=1 \\ y=1 \end{cases}$$

distanta $d(v, U) =$

$= d(v, \operatorname{pr}_U v) =$

$= \| v - \operatorname{pr}_U v \|$



$$\|v\|^2 = \| \underbrace{\operatorname{pr}_U v}_{\in} + \underbrace{\operatorname{pr}_{U^\perp} v}_{\perp} \|^2 = \|\operatorname{pr}_U v\|^2 + \underbrace{\|\operatorname{pr}_{U^\perp} v\|^2}_{\geq 0} \geq$$

$$\geq \|\operatorname{pr}_U v\|^2$$

$$v - \operatorname{pr}_{\ker(A)} v = \operatorname{pr}_{\ker(A)^\perp} v = \operatorname{pr}_{\operatorname{Im}(A^t)} v$$

Least squares.

$$\mathbb{R}^n = \sqrt{(\cdot)^2 + (\cdot)^2 + \dots (\cdot)^2}$$